GRAHITH G RAO

AI / Research Engineer | LLM Agents & RAG | Computer Vision & Multimodal AI

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Scan for portfolio

SUMMARY

AI/ML engineering student (B.E., 2027) who builds **production-grade LLM agents, RAG systems and sensor-to-speech hardware**. Ships tested, containerised, documented code with a focus on reliability, grounding and access control. Seeking AI / Research Engineer roles in agents, computer vision and multimodal AI.

PROJECTS

MCP Tool-Calling AI Agent | Secure enterprise agent

github.com/grahithrao/MCP-Tool-Calling-AI-Agent

Python, LangGraph, Model Context Protocol, FastAPI, ChromaDB, React

- Engineered a LangGraph agent that discovers **11 tools across 4 MCP servers** at runtime, so new capabilities need no prompt or agent changes.
- Placed access control **outside the model** (role-based permissions, tool allow-listing, scope-filtered retrieval, audit log) and verified it with **134 automated tests**.

RAG From Scratch | Grounded document Q&A

github.com/grahithrao/rag-from-scratch

Python, FastAPI, ChromaDB, Sentence-Transformers, Streamlit, Docker, GitHub Actions

- Built an end-to-end RAG system (parsing, chunking, **local 384-d embeddings**, ChromaDB retrieval) behind a FastAPI backend and Streamlit interface.
- Grounded every answer in retrieved passages with **exact citations** and a refusal when evidence is missing; containerised with Docker and CI.

PostPilot | AI agent with human approval gate

github.com/grahithrao/postpilot

TypeScript, LangGraph, Prisma, PostgreSQL, Zod, OAuth

- Built a LangGraph agent that drafts and scores posts, then **pauses for human approval**; state is checkpointed to Postgres so runs resume after restarts.
- Enforced "no publish without approval" in **4 independent layers**, with at-most-once publishing, retries, encrypted OAuth tokens and **103 test assertions**.

SignSpeak Gloves | Sign language to speech

github.com/grahithrao/signspeak

Arduino, C++, HC-05 Bluetooth, React Native (Expo), TypeScript, Java

- Built a sensor glove (Arduino, **15 channels over Bluetooth**) and an Android app that converts hand gestures into speech.
- Designed the recognition pipeline (calibration, filtering, **cosine-similarity matching**, state machine) with trainable gestures and a hardware-free simulator.

Also built: RoadSaathi (roadside-assistance marketplace) | Bulk Certificate Generator (async job queue)

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, C/C++, Java
LLM & Agents	LangGraph, MCP, RAG, tool calling
ML & Vision	TensorFlow, Scikit-learn, OpenCV, NLP
Backend & DevOps	FastAPI, NestJS, Next.js, React, PostgreSQL, Prisma, Docker
Hardware & Cloud	Arduino, sensor interfacing, Bluetooth (HC-05), AWS

EDUCATION

B.E. in Artificial Intelligence & Machine Learning, AJ Institute of Engineering & Technology — 2023 - 2027 (Expected)

PUC, Science (PCMC), Canara PU College — 2021 - 2023

CERTIFICATIONS & LEADERSHIP

IBM: Machine Learning with Python (2024) | IBM: Generative AI Fundamentals (2024)
AWS Cloud Computing Foundations (2025) | NPTEL (IIT): Intro to ML (2023) | Deloitte Analytics Simulation (2026)
Leadership: Led the final-year major project team | **Service:** NSS volunteer, beach clean-up drives